

PYTHON FUNCTIONS AND EXPRESSIONS Solutions

COMPUTER SCIENCE MENTORS 61A

September 2 – September 5, 2025

1 Expressions

1. What Would Python Display?

```
>>> 3

3

>>> "cs61a"

'cs61a'

>>> x = 3
>>> x

3

>>> x = print("cs61a")
cs61a
>>> x

>>> print(print(print("cs61a")))

cs61a
None
None

>>> def f1(x):
...     return x + 1
>>> f1(3)

4

>>> f1(2) + f1(2 + 3)

9
```

```
>>> def f2(y):  
...     return y / 0  
>>> f2(4)
```

ZeroDivisionError: division by zero

```
>>> def f3(x, y):  
...     if x > y:  
...         return x  
...     elif x == y:  
...         return x + y  
...     else:  
...         return y  
>>> f3(1, 2)
```

2

```
>>> f3(5, 5)
```

10

```
>>> 1 or 2 or 3
```

1

```
>>> 1 or 0 or 3
```

1

```
>>> 4 and (2 or 1/0)
```

2

```
>>> 0 or (not 1 and 3)
```

False

```
>>> (2 or 1/0) and (False or (True and (0 or 1)))
```

1

2. What would Python display for each expression?

(a) $3 + 4$

7

(b) `'3' + 4`

TypeError

(c) `'3 + 4'`

`'3 + 4'`

(d) `'3' + '4'`

`'34'`

2 Functions

1. For the following expressions, simplify the operands in the order of evaluation of the entire expression

Example: `add(3, mul(4, 5))`

Order of Evaluation: `add(3, mul(4, 5))` $\rightarrow$ `add(3, 20)` $\rightarrow$ 23

(a) `add(1, mul(2, 3))`

```
add(1, mul(2, 3))
add(1, 6)
7
```

(b) `add(mul(2, 3), add(1, 4))`

```
add(mul(2, 3), add(1, 4))
add(6, add(1, 4))
add(6, 5)
11
```

(c) `max(mul(1, 2), add(5, 6), 3, mul(mul(3, 4), 1), 7)`

```
max(mul(1, 2), add(5, 6), 3, mul(mul(3, 4), 1), 7)
max(2, add(5, 6), 3, mul(mul(3, 4), 1), 7)
max(2, 11, 3, mul(mul(3, 4), 1), 7)
max(2, 11, 3, mul(12, 1), 7)
max(2, 11, 3, 12, 7)
12
```

2. Write a function `fahrenheitToCelsius`, which takes in an integer `f` and returns the temperature in Celsius. Hint: the formula for Fahrenheit to Celcius is to subtract 32 and then multiply by 5/9.

```
def fahrenheitToCelsius(f):  
    """  
    >>> fahrenheitToCelsius(32)  
    0.0  
    >>> fahrenheitToCelsius(104)  
    40.0  
    """
```

```
def fahrenheitToCelsius(f):  
    return (f - 32) * (5/9)
```

3. Sally really wants to know what she needs to score to pass her class! Fill out the function `score_needed` that takes in `current_grade`, her current grade, `num_exams`, the number of exams taken so far (not including her final exam), and `pass_grade`, the grade needed to pass. Assume that every exam, including the final exam, has the same weight.

```
def score_needed(current_grade, num_exams, pass_grade):  
    """  
    >>> score_needed(89, 4, 90)  
    94  
    >>> score_needed(65, 2, 70)  
    80  
    >>> score_needed(77, 10, 78)  
    88  
    """
```

```
def score_needed(current_grade, num_exams, pass_grade):  
    total_pts = current_grade * num_exams  
    total_needed = pass_grade * (num_exams + 1)  
    return total_needed - total_pts
```

4. What would Python display?

```
>>> from operator import sub, mul
>>> def print_sub(x, y):
...     print('sub')
...     return sub(x, y)
>>> def print_mul(x, y):
...     print('mul')
...     return mul(x, y)
>>> print_sub(print_mul(505, 4), 3)
```

```
mul
sub
2017
```